

Introduction to GPS

Learning Objectives

- Define GPS and explain how it works
- Identify devices that use GPS technology
- Describe real-life applications of GPS

Engage (5-7 minutes)

When a bus driver travels from your village to the city, he follows road signs and landmarks to know where he is. GPS is like invisible road signs in the sky that tell devices exactly where they are. Let us explore this amazing technology.

Explore (10-12 minutes)

Look at a physical map of your district. Mark your village, the nearest town, and the district headquarters. Now think about how a phone knows which of these three places you are at. Write your ideas on paper and share with the class.

Explain (10-12 minutes)

GPS stands for Global Positioning System. It uses satellites orbiting Earth to send signals to devices like phones and car navigation systems. By receiving signals from multiple satellites, a device calculates its exact location—latitude, longitude, and altitude.

Elaborate (8-10 minutes)

GPS helps in many ways: auto-rickshaws find addresses, delivery trucks track routes, farmers use precision farming, and ambulances reach patients faster. Write three ways GPS could help people in your village specifically, like finding lost cattle or guiding emergency vehicles.

Evaluate (5-8 minutes)

Exit ticket: Write the full form of GPS and explain in one sentence how it works. Name one device that uses GPS.